

Notes: Stevens (RES, 2020)

Coarse Pricing Policies

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Nielsen Retail Scanner data	2
2.2	Break test and policies	2
2.3	Policy types	2
3	Models and empirical specifications	2
3.1	Firm objective	2
3.2	Rational inattention and discrete prices	3
3.3	Inflation decomposition	3
4	Identification and empirical design	3
5	Section-by-section study guide	3
6	Tables and figures	3
6.1	Table 1: Characteristics of pricing policies + Figure 1: Frequency and size of within and across policy price changes across product groups	3
6.2	Figure 2: Classification by policy type + Table 2: Price discrimination policies	4
6.3	Figure 3 and Table 3: Inflation dynamics by policy type	4
6.4	Figure 4 and Figure 5: Great Recession policy adjustment	5
6.5	Figure 6 and Table 4: Model timing and baseline parameterization	5
6.6	Figure 7 and Table 5: Pricing policies in the model	6
6.7	Figure 8 and Table 6: Model outcomes and alternative MRP parameterizations	6
6.8	Figure 9: Growth of new mass points in the price distribution	7
6.9	Figure 10 and Table 7: Impulse responses and higher volatility	7
7	Short summary	8
8	Expanded conclusion	8
8.1	Core conclusion	8
8.2	Economic intuition	8
8.3	Great Recession implication	8
8.4	Monetary policy implication	8
8.5	Limitations	9
8.6	Takeaway	9

1 Big picture

- **Question.** Why did inflation respond mildly during the Great Recession despite large changes in activity and high micro price volatility?
- **Main empirical claim.** Firms use coarse pricing policies: menus of a few prices that are updated infrequently.
- **Core fact.** Prices can change often inside a policy, but the set of available price levels changes slowly.
- **Policy types.** Products are classified as single-price (SPP), one-to-flex (OFP), and multi-rigid (MRP).
- **Theory.** Costly information makes firms approximate a continuously moving target price with a small menu.
- **Aggregate implication.** High micro price volatility does not imply high aggregate price flexibility.

2 Data and measurement

2.1 Nielsen Retail Scanner data

The paper uses weekly store-UPC prices from the Nielsen Retail Scanner Database, 2006–2015. The data contain more than one million store-UPC series and are high-frequency enough to study within-policy price changes.

Interpretation: High-frequency data are essential because the paper distinguishes changes inside a pricing policy from changes in the policy itself.

2.2 Break test and policies

Pricing policies are identified with an adapted Kolmogorov–Smirnov break test:

$$D(\tau) = \sup_p |F_{1:\tau}(p) - F_{\tau+1:T}(p)|.$$

The method detects changes in the full distribution of prices, not just changes in a modal or reference price.

Interpretation: The break test lets the data reveal stable price menus.

2.3 Policy types

SPP series have one sticky price; OFP series have one rigid price plus flexible deviations; MRP series have multiple rigid prices. MRP products are the dominant and most volatile category.

Interpretation: The categories describe the structure of pricing plans, not merely raw price-change frequencies.

3 Models and empirical specifications

3.1 Firm objective

$$\max E_0 \sum_{t=0}^{\infty} \beta^t [\pi_{it} - \theta_p I_{it}^p - \theta_r I_{it}^r - \kappa \delta_{it}^r].$$

The firm pays for a price signal, a review signal, and a fixed cost of policy review.

Interpretation: The firm faces two information problems: when to redesign the price menu and which price in the current menu to charge.

3.2 Rational inattention and discrete prices

Information cost is proportional to entropy reduction. A continuous target price is mapped into a finite set of possible price choices.

Interpretation: Discrete price menus are not assumed; they arise as optimal coarse approximations to a continuous target.

3.3 Inflation decomposition

$$\pi_t = \sum_g w_{g,t} \pi_{g,t}.$$

Inflation is decomposed by policy type and by across-policy versus within-policy price changes.

Interpretation: The decomposition links micro policy structure to aggregate inflation.

4 Identification and empirical design

The paper is mainly a measurement and quantitative-modeling exercise. It identifies pricing policies from distributional breaks, classifies products by policy structure, uses Great Recession variation to study cyclical adjustment, and calibrates the rational-inattention model to micro moments. The main limitation is that internal firm information acquisition is not observed directly.

5 Section-by-section study guide

- **Introduction.** Motivates the muted inflation puzzle and argues for information frictions.
- **Empirical section.** Documents coarse policies, policy types, and recession dynamics.
- **Theory section.** Builds a model with review and pricing signals.
- **Numerical section.** Shows the model matches micro facts and generates sluggish aggregate price responses.

6 Tables and figures

6.1 Table 1: Characteristics of pricing policies + Figure 1: Frequency and size of within and across policy price changes across product groups

Read: Table 1 reports SPP 12.0%, OFP 28.5%, and MRP 59.5%. MRP products have the highest within-policy price-change frequency and cardinality. Figure 1 shows positive frequency-size patterns for within-policy and across-policy changes.

Detailed interpretation: These visuals establish that the relevant object is a sticky policy menu, not an individual price.

Economic message: Frequent price changes can coexist with price-level rigidity.

TABLE 1
Characteristics of pricing policies

	All	Single-price	One-to-flex	Multi-rigid
Fraction of series (%)	100	12.0	28.5	59.5
Monthly frequency of policy changes (%)	12.2	7.8	17.3	12.0
Implied policy duration (months)	7.7	12.3	5.3	7.9
Frequency of weekly price changes within (%)	24.6	0.6	15.0	35.8
Size of price changes within (%)	11.9	5.8	9.9	13.6
Size of shift across (%)	11.3	8.5	11.7	11.5
Policy cardinality	3	1	3	4

Note: Nielsen Retail Scanner data, 2006–15. *Implied policy duration* is the duration implied by the median monthly frequency of policy changes. *Frequency of weekly price changes within* is the median weekly frequency with which prices change between breaks. *Size of price changes within* is the absolute value and is non-zero for SPP because the category allows for rare deviations from the modal price. *Size of shift across* is the median absolute change in the weighted average price across different policy realizations.

(a) Table 1: Characteristics of pricing policies

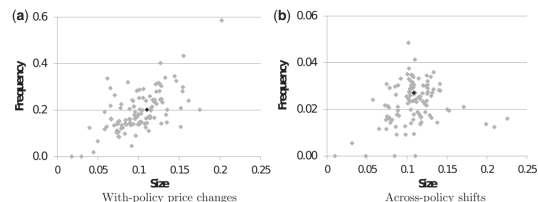


FIGURE 1
Frequency and size of within and across policy price changes across product groups

Note: Nielsen Retail Scanner data, 2006–15. (a) (With-policy price changes) The frequency against the absolute size of within-policy price changes. (b) (Across-policy shifts) The frequency of policy changes against the absolute size of the shift in the average price across consecutive policies. Points indicate means at the product group level. The expenditure-weighted medians for the full sample are in black.

(b) Figure 1: Frequency and size of price changes

6.2 Figure 2: Classification by policy type + Table 2: Price discrimination policies

Read: MRP products are widespread across groups. Table 2 shows that price discrimination explains part, but not all, of OFP and MRP volatility.

Detailed interpretation: Non-PD MRP products remain highly volatile and coarse.

Economic message: Coarse multi-price policies are broader than sale pricing.

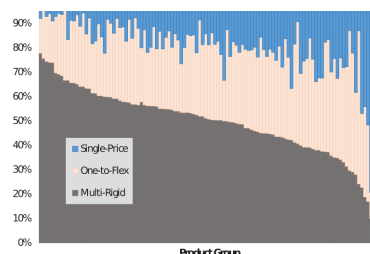


FIGURE 2

Classification of series by type of pricing policy, across product groups.

Note: Nielsen Retail Scanner data, 2006–15. Percent of series of each type in each product group.

(a) Figure 2: Classification by policy type

TABLE 2
Price discrimination policies

	One-to-flex		Multi-rigid	
	PD	Non-PD	PD	Non-PD
Fraction of all series (%)	10.2	18.3	29.5	30.0
Monthly frequency of policy changes (%)	11.6	21.3	9.1	14.8
Implied policy duration (months)	8.1	4.2	10.4	6.2
Frequency of weekly price changes within (%)	14.9	15.0	33.3	38.6
Size of price changes within (%)	11.8	9.1	15.3	12.2
Size of policy shift (%)	11.9	12.6	12.4	10.4
Policy cardinality	3	2	4	4

Note: Nielsen Retail Scanner data. Statistics for one-to-flex and multi-rigid series that are *PD* and *non-PD* series, where *PD* series are defined as series in which a majority of policy realizations have the maximum price equal to the modal price.

(b) Table 2: Price discrimination policies

6.3 Figure 3 and Table 3: Inflation dynamics by policy type

Read: Figure 3 compares Nielsen inflation, CPI inflation, crude oil, and policy-type inflation. Table 3 estimates sensitivity of inflation to unemployment by policy type.

Detailed interpretation: MRP products dominate inflation volatility.

Economic message: Aggregate inflation depends on the composition of micro pricing policies.

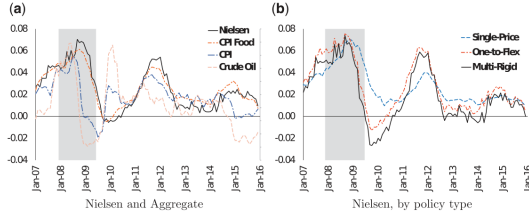


FIGURE 3

Annualized inflation in Nielsen and the CPI versus crude oil

Note: Nielsen Retail Scanner and Bureau of Labor Statistics data. Crude oil inflation is re-scaled for comparability. The shading marks the Great Recession. (a) Nielsen and Aggregate data. (b) Nielsen, by policy type.

(a) Figure 3: Nielsen and CPI inflation versus crude oil

	National		State-level	
	(1)	(2)	(3)	(4)
Unemployment	-0.046 (0.038)	-	0.015 (0.041)	-
Unemployment $\times$ OFP	-0.042 (0.050)	-0.089 (0.049)	-0.071** (0.027)	-0.075*** (0.027)
Unemployment $\times$ MRP	-0.098* (0.047)	-0.146** (0.045)	-0.094*** (0.026)	-0.099*** (0.027)
Module FE	Yes	-	-	-
Module $\times$ Month FE	-	Yes	Yes	-
State FE	-	-	-	Yes
State $\times$ Module $\times$ Month FE	-	-	-	-
Observations	254,600	244,820	9,626,039	9,079,654
R ²	0.0608	0.5304	0.2148	0.5669

Note: Nielsen Retail Scanner data, 2006–15. The first two columns report regressions of module-policy inflation on national unemployment. Standard errors are in parentheses, clustered at the module level. Two-way clustering by module and month (omitted) reduces significance from the 1% to the 5% level for MRP series. The last two columns report regressions of module-policy inflation by state on state-level unemployment as a proxy for local demand. Standard errors are in parentheses, two-way clustered at the state and month level. All regressions also include policy dummies and fixed effects, as indicated (not reported). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

(b) Table 3: Sensitivity of inflation to unemployment

6.4 Figure 4 and Figure 5: Great Recession policy adjustment

Read: Figure 4 shows policy choice over time and contributions to inflation. Figure 5 shows that the recession increased policy-change frequency much more than within-policy flexibility.

Detailed interpretation: Firms react to uncertainty mainly by revising policies more often.

Economic message: High micro volatility does not mechanically imply flexible aggregate prices.

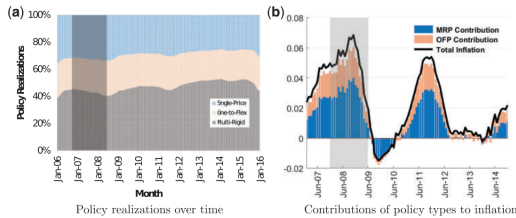


FIGURE 4

Policy choices and their contribution to inflation over time

Note: Nielsen Retail Scanner and Bureau of Labor Statistics data. (a) The type of policy realizations over time. (b) The contribution of MRP and OFP series to total inflation. The difference between total inflation (marked by the black line) and the sum of MRP and OFP contributions is the contribution of SPP series.

(a) Figure 4: Policy choices and inflation contribution

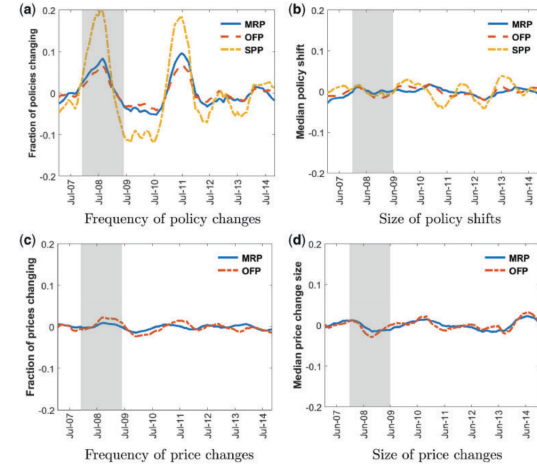


FIGURE 5

Policy and price adjustment during the Great Recession

Note: Nielsen Retail Scanner Data. The top panels show time series for the deviations from the trend in the fraction of policy changes over time (a) and in the median absolute shift in prices across policies (b), by policy type. The shift is obtained by computing the change across policies in the weighted average price. The bottom panels show time series for the frequency of within-policy price changes (c) and for the size of within-policy price changes (d), by policy type. These are seasonally adjusted weekly figures, averaged to monthly values and filtered with a Baxter-King bandpass filter with parameters 12, 96, and 18.

(b) Figure 5: Policy and price adjustment in the Great Recession

6.5 Figure 6 and Table 4: Model timing and baseline parameterization

Read: Figure 6 shows the review and pricing sequence. Table 4 reports the model parameters.

Detailed interpretation: The model has two costly information choices: policy review and price selection.

Economic message: Coarse pricing is a result of costly information acquisition.

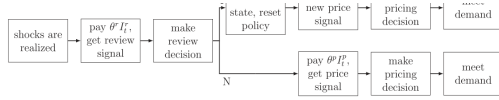


FIGURE 6
Sequence of events in each period of the model

(a) Figure 6: Sequence of events

TABLE 4
Baseline parameterization

Parameter	Symbol	Values	Explanation/target
Discount factor	β	0.9994	Annual discount rate of 3%
Elasticity of substitution	ε	5	Full info markup of 25%
Elast. of inter-temporal subst.	σ	2.7	Strategic complementarities
Inverse Frisch elasticity	ν	0	Indivisible labour
Inverse production fn. exponent	γ	1.5	Decreasing returns to scale
Fixed cost of policy review	κ	1.65; 1.8	Frequency of policy reviews
Cost of review signal	θ^P	4	Price shift across policies
Cost of price signal	θ^P	>0.13; 0.1	Cardinality of policy
Std. dev. of idio. quality shock	σ_ε	0.016; 0.028	Size of price changes

Note: Where there are two values, the first indicates the SPP parameterization, and the second indicates the MRP parameterization.

(b) Table 4: Baseline parameterization

6.6 Figure 7 and Table 5: Pricing policies in the model

Read: The model matches SPP and MRP duration, cardinality, within-policy volatility, information expenditure, and overpricing.

Detailed interpretation: MRP firms spend more on information and still charge prices above full-information prices.

Economic message: Information frictions generate realistic coarse price plans.

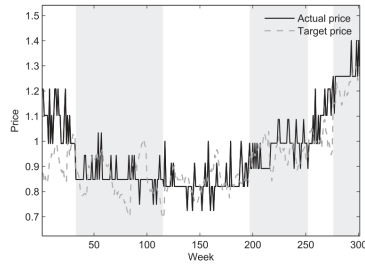


FIGURE 7
Simulated price series
Note: Shading marks the policy reviews identified by the break test.

(a) Figure 7: Simulated price series

TABLE 5
Pricing policies in the model

	Single-price		Multi-rigid (non-PD)	
	Data	Model	Data	Model
Targets				
Cardinality of the pricing policy	1	1	4	4
Weekly frequency of policy reviews (%)	1.8	1.8	3.4	3.4
Shift in prices across policies (%)	8.5	8.5	10.4	10.8
Weekly frequency of modal price (%)	100	100	58.1	66.3
Weekly frequency of price changes within (%)	—	—	38.6	39.4
Size of price changes within (%)	—	—	12.2	11.1
Information expenditure (% of revenues)				
On reviews		1.8		3.0
On review signal		0.7		0.6
On price signal		0.0		1.6
Total info expenditure		2.5		5.2
Profits, excluding info costs (% FI)		91.2		90.0
Threshold cost θ_p for acquiring price signal		0.13		0.42
Amount by which prices exceed FI price (%)		2.9		4.5

Note: Data versus baseline model results. Profits are reported as a percentage of the full information (FI) profits.

(b) Table 5: Pricing policies in the model

6.7 Figure 8 and Table 6: Model outcomes and alternative MRP parameterizations

Read: Figure 8 shows pre-review targets, review hazards, and post-review distributions. Table 6 changes information costs and volatility.

Detailed interpretation: Review policy and price-menu complexity interact.

Economic message: Firms substitute between better price information and more frequent reviews.

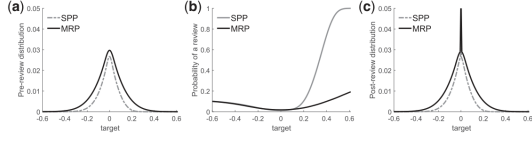


Figure 8: Model outcomes. (a) Distribution of pre-review targets. (b) Hazard function for policy reviews. (c) Distribution of targets post the review decision. The figures correspond to the SPP and MRP parameterizations shown in Table 5.

(a) Figure 8: Model outcomes

	Base	High θ^P	High θ^T	High κ
Polices				
Cardinality of the pricing policy	4	1	5	5
Weekly frequency of policy reviews (%)	3.4	2.8	4.4	2.8
Shift in prices across policies (%)	10.8	12.9	9.3	11.4
Information expenditure (% of revenues)				
On reviews	3.0	3.3	4.7	3.2
On review signal	0.6	1.4	1.7	0.9
On price signal	1.6	—	1.5	2.8
Total info expenditure	5.2	4.6	6.4	6.9
Profits, excluding info costs (%FI)	90.0	85.2	90.0	89.1
Amount by which prices exceed FI price (%)	4.5	4.9	4.5	5.0

Notes: The first column shows the baseline MRP parameterization. Each subsequent column considers a single parameter change: $\theta^P = 0.42$, which is the threshold information cost for multiple-price policies; $\theta^T = 20$, which generates a near-constant probability of policy reviews; and $\kappa = 3.6$, which also generates a very flat hazard function for policy reviews.

(b) Table 6: Alternative parameterizations

6.8 Figure 9: Growth of new mass points in the price distribution

Read: As price information becomes cheaper, new price points appear in the optimal pricing support.

Detailed interpretation: Discrete menus arise endogenously even with continuous target-price shocks.

Economic message: Information costs create lumpy price supports.

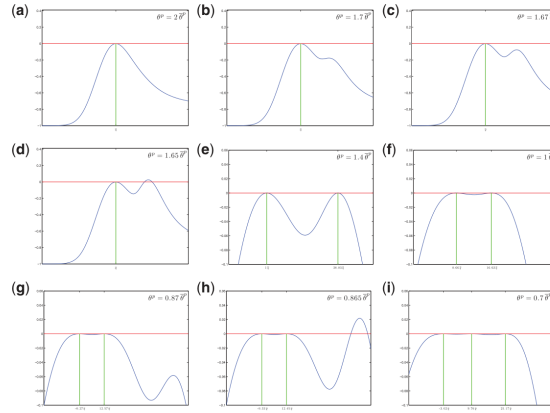


Figure 9: Growth of new mass points in the price distribution. The panels plot the function $Z(q) - 1$ as the cost of information θ^P is reduced. The points of support are shown as multiples of $\bar{q}$, the price that would be charged under the SPP.

Figure 9: Growth of new mass points

6.9 Figure 10 and Table 7: Impulse responses and higher volatility

Read: Figure 10 compares price-index impulse responses across specifications. Table 7 studies a 10% increase in fundamental volatility.

Detailed interpretation: Higher volatility raises information expenditure and precautionary overpricing but does not eliminate sluggish adjustment.

Economic message: Uncertainty can increase real rigidity and monetary policy effectiveness.

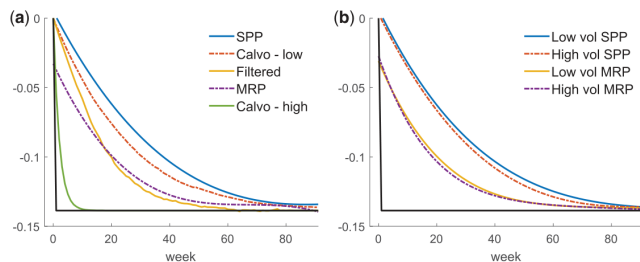


FIGURE 10

Impulse response functions across model specifications

Note: Model results. (a) The impulse response functions of the price index to a negative nominal demand shock for the baseline SPP and MRP series, as well as for three alternative parameterizations: "Filtered" shows the response of the SPP model calibrated to match the frequency of policy reviews and the shift across policies seen in the MRP data; "Calvo-low" is the response of the standard Calvo model calibrated to match these same statistics; and "Calvo-high" is the response of the standard Calvo model calibrated to match the frequency and size of all price changes in the MRP data. In all cases the MRP statistics are for the non-PD series. (b) The impulse response functions of the price index to the same shock in a low versus high volatility environment, for both the SPP and MRP models.

(a) Figure 10: Impulse response functions

TABLE 7
The effects of an increase in fundamental volatility

	MRP series	SPP series
Change in frequency of policy reviews (%)	7.8	9.1
Change in shift across policies (%)	6.6	4.3
Change in frequency of price changes within (%)	3.0	
Change in size of price changes within (%)	1.1	
Change in total spending on information (%)	10.3	10.8
Change in profits relative to FT (ex-ante) (%)	-0.7	-0.8
Change in average prices charged (%)	0.5	0.3

Note: Model results. The table shows changes in key statistics as a result of a 10% increase in volatility relative to the baseline parameterizations.

(b) Table 7: Effects of increased volatility

7 Short summary

- Firms use coarse pricing policies: small price menus updated infrequently.
- MRP products dominate the sample and contribute most to inflation volatility.
- The break test separates within-policy price changes from across-policy policy changes.
- During the Great Recession, firms changed policies more often but did not make within-policy prices much more flexible.
- A rational-inattention model matches the micro facts and implies sluggish aggregate price responses.

8 Expanded conclusion

8.1 Core conclusion

Product-level price changes are frequent, but pricing policies are sticky and coarse. Aggregate rigidity arises because firms move inside menus rather than continuously reoptimizing.

8.2 Economic intuition

Firms face continuous target-price movements but costly information. A small price menu is a cheap approximation to a fully informed price rule.

8.3 Great Recession implication

In uncertain times, firms update policies more often and overprice more cautiously, helping explain muted disinflation.

8.4 Monetary policy implication

Information frictions can make prices sluggish even when micro price changes are frequent, increasing the real effects of demand shocks.

8.5 Limitations

The scanner data do not cover the whole CPI basket, and the model abstracts from many retail details such as inventory, multi-product pricing, and strategic sales.

8.6 Takeaway

Price flexibility must be measured at the level of pricing policies, not merely at the level of posted price changes.